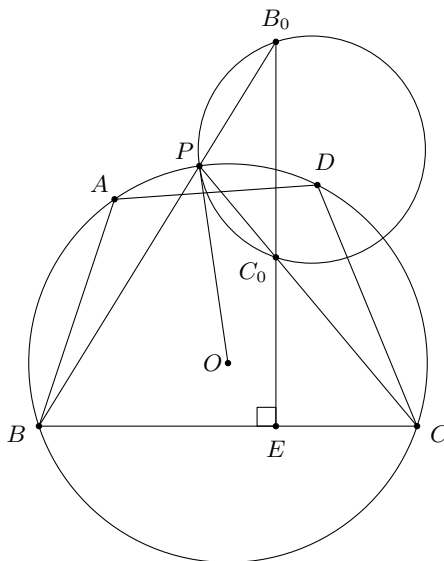


Problem 57. Let $ABCD$ be a cyclic quadrilateral. A point P moves along the arc AD which does not contain B and C . A fixed line ℓ , perpendicular to BC , meets the rays BP, CP at points B_0, C_0 respectively. Prove that the tangent at P to the circumcircle of triangle PB_0C_0 passes through some fixed point.



Solution. We only work on the configuration as shown. The other cases are similar. Let ℓ meet BC at E . Let O be the centre of $(ABCD)$. Then

$$\angle OPC_0 = \angle OPC = 90^\circ - \angle CBP = \angle PB_0C_0.$$

This shows OP is tangent to (PC_0B_0) . Thus, the fixed point O always lie on the tangent to (PC_0B_0) at P . $\square$

Problem 58. Let n be a positive integer. n people take part in a certain party. For any pair of the participants, either the two are acquainted with each other or they are not. What is the maximum possible number of the pairs for which the two are not acquainted but have a common acquaintance among the participants?

Solution. We use terminologies in graph theory. We call a pair of vertices good if they are not adjacent and they share a common neighbour. Consider the following example. One of the vertices A is adjacent to all other vertices, and no two other vertices are adjacent. Then A cannot form any good pair with the others by definition, but all other pairs of vertices are good. There are $\binom{n-1}{2} = \frac{(n-1)(n-2)}{2}$ good pairs in this case.

If there are at least $n-1$ edges, then there are $n-1$ pairs of vertices which are the endpoints of these edges. These pairs of vertices are not good by definition. So the number of good pairs is at most $\binom{n}{2} - (n-1) = \frac{(n-1)(n-2)}{2}$.

If there are at most $n-2$ edges, then there are at most $\binom{n-2}{2}$ good pairs since each good pair A, B corresponds to a pair of edges AC, BC , and each pair of edges corresponds to at most one good pair.

In both cases, the number of good pairs cannot exceed $\frac{(n-1)(n-2)}{2}$. From the initial construction, this is the maximum possible number of good pairs. $\square$

Problem 59. Let n be an odd positive integer, and let $x_1, x_2, \dots, x_n$ be non-negative real numbers. Show that

$$\min_{1 \leq i \leq n} (x_i^2 + x_{i+1}^2) \leq \max_{1 \leq j \leq n} (2x_j x_{j+1})$$

where $x_{n+1} = x_1$.

Solution. Suppose on the contrary that

$$\min_{1 \leq i \leq n} (x_i^2 + x_{i+1}^2) > \max_{1 \leq j \leq n} (2x_j x_{j+1}).$$

This means $x_i^2 + x_{i+1}^2 > 2x_j x_{j+1}$ for any $1 \leq i, j \leq n$. By putting $i = j$, we deduce $(x_i - x_{i+1})^2 > 0$. This implies $x_i \neq x_{i+1}$. If $x_i < x_{i+1}$, then from $x_i^2 + x_{i+1}^2 > 2x_{i+1}x_{i+2}$ we have

$$2x_{i+1}x_{i+2} < x_i^2 + x_{i+1}^2 < 2x_{i+1}^2.$$

Note that $x_{i+1} > 0$ as $x_{i+1} > x_i \geq 0$. So this yields $x_{i+2} < x_{i+1}$.

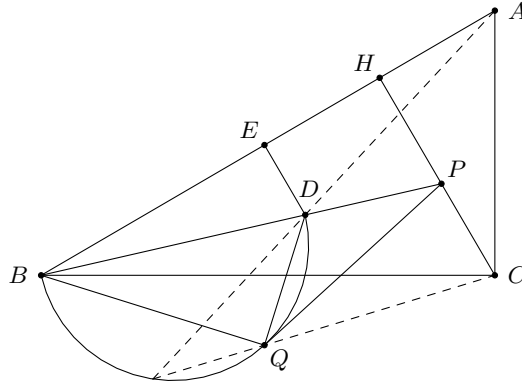
If $x_i > x_{i+1}$, then from $x_{i+1}^2 + x_{i+2}^2 > 2x_i x_{i+1}$ we have

$$x_{i+1}^2 + x_{i+2}^2 > 2x_i x_{i+1} \geq 2x_{i+1}^2.$$

This yields $x_{i+2}^2 > x_{i+1}^2$ and hence $x_{i+2} > x_{i+1}$.

WLOG assume $x_1 > x_2$. Then the above results show that $x_2 < x_3$, $x_3 > x_4$, $x_4 < x_5$ and so on. Since n is odd, inductively this implies $x_n > x_1$ and $x_1 < x_2$. This is a contradiction. Therefore, the inequality must hold. $\square$

Problem 60. Let ABC be a triangle with $\angle C = 90^\circ$, and let H be the foot of the altitude from C . A point D is chosen inside the triangle CBH so that CH bisects AD . Let P be the intersection point of the lines BD and CH . Let ω be the semicircle with diameter BD that meets the segment CB at an interior point. A line through P is tangent to ω at Q . Prove that the lines CQ and AD meet on ω .



Solution. Let E be the projection of D on AB . As CH bisects AD , H is the midpoint of AE . As BD is the diameter of ω , we have $\angle DQB = 90^\circ = \angle ACB$. From $\triangle PDQ \sim \triangle PQB$, we obtain

$$\frac{DQ}{QB} = \frac{PD}{PQ} = \frac{PQ}{PB}.$$

Therefore,

$$\frac{DQ}{QB} = \sqrt{\frac{PD}{PQ} \times \frac{PQ}{PB}} = \sqrt{\frac{PD}{PB}} = \sqrt{\frac{HE}{HB}} = \sqrt{\frac{HA}{HB}}.$$

As $\angle ACB = \angle CHA = 90^\circ$, we have $\triangle CHA \sim \triangle BHC$. This yields

$$\frac{CA}{BC} = \frac{CH}{BH} = \frac{HA}{HC}.$$

Thus,

$$\frac{CA}{BC} = \sqrt{\frac{CH}{BH} \times \frac{HA}{HC}} = \sqrt{\frac{HA}{HB}} = \frac{DQ}{QB}.$$

Together with $\angle DQB = 90^\circ = \angle ACB$, we find that $\triangle DBQ \sim \triangle ABC$. By spiral similarity, AD and CQ meet on (BDQ) . From the configuration, this intersection point must lie on ω . $\square$